

Arvind Murugan

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Appointments

Professor

PHYSICS + JAMES FRANCK INSTITUTE

University of Chicago

2026 -

Associate Professor

PHYSICS + JAMES FRANCK INSTITUTE

University of Chicago

2023 - 2026

Assistant Professor

PHYSICS + JAMES FRANCK INSTITUTE

University of Chicago

2015 - 2023

Postdoctoral fellow

APPLIED MATH (MICHAEL BRENNER GROUP)

Harvard University

2012 - 2015

Visiting researcher

STANISLAS LEIBLER GROUP

Rockefeller University

2009 - 2012

Member

SIMONS CENTER FOR SYSTEMS BIOLOGY

Institute for Advanced Studies

2009 - 2012

Education

Princeton University

PH.D IN PHYSICS

Princeton, NJ

09/2004 - 07/2009

- Thesis: Gauge-gravity duality with renormalization group flow and reduced supersymmetry
- Advisor: Igor R. Klebanov

California Institute of Technology

B.S. IN MATHEMATICS W/ HONORS

Pasadena, CA

09/2000 - 06/2004

- Advisor: Anton Kapustin

Awards

- 2025 **Schmidt Polymath Award**, Schmidt Sciences
- 2025 **International Prize in Biophysics**, Tel Aviv
- 2023 **National Science Foundation CAREER Award**, NSF
- 2017 **Simons Investigator**, Simons Foundation
- 2010 **Addie and Harold Broitman Member in Biology**, Institute for Advanced Study
- 2004 **Princeton University Graduate Centennial Fellowship**, Princeton University

Service

2025 - 28	Advisory Board , Kavli Institute for Theoretical Physics	<i>Santa Barbara, CA</i>
2023 - 28	Deputy Director , U Chicago Center for Living Systems, NSF Physics Frontier Center	<i>Chicago, IL</i>
2026	Organizer , KITP summer workshop	<i>Santa Barbara, CA</i>
2025 - 27	Chair line , Gordon Conference Stochastic Physics in Biology	<i>Ventura, CA</i>
2024 - 25	Chair line , APS Delbruck prize committee	
2025	Planning committee , National Academies Workshop on Frontiers of Living Materials	<i>Washington D.C</i>
2025	Panelist , NSF Panel Review	<i>Alexandria, VA</i>
2025	Site Visit Panel , NSF Panel Review	<i>Alexandria, VA</i>
2024	Organizer , Aspen Institute for Physics, Winter workshop	<i>Aspen, CO</i>
2024	Panelist , NSF Panel Review	<i>Alexandria, VA</i>
2024	Selection committee , New IBD chair, UChicago	<i>Chicago, IL</i>
2022 -	Reviewing Editor , eLife	
2018 - 21	Program committee , Annual q-Bio meeting	
2017 - 23	Organizer , APS March meeting sessions	

Teaching

Ph 490	Principles of Biological Physics	<i>Fall 2015</i>
Ph 330	Mathematical Methods for Physicists	<i>Fall 2016</i>
Ph 490	Principles of Biological Physics	<i>Fall 2017</i>
Ph 330	Mathematical Methods for Physicists	<i>Fall 2017</i>
Ph 255	Biological Physics	<i>Spring 2018</i>
Ph 330	Mathematical Methods for Physicists	<i>Winter 2019</i>
Ph 121	General Physics	<i>Fall 2019</i>
Ph 330	Mathematical Methods for Physicists	<i>Winter 2020</i>
Ph 121	General Physics	<i>Fall 2020</i>
Ph 121	General Physics	<i>Fall 2021</i>
Ph 490	Biological Physics	<i>Fall 2022</i>
Ph 121	General Physics	<i>Fall 2023</i>
Ph 255	Biological Physics	<i>Winter 2023</i>
Ph 121	General Physics	<i>Fall 2024</i>
Ph 490	Biological Physics	<i>Winter 2024</i>
Ph 131	General Physics	<i>Fall 2025</i>

Support

2025 - 30	Schmidt Sciences , Schmidt Polymath Award - PI
2023 - 28	National Institute of Health , R35 award - PI
2023 - 27	National Science Foundation , Physics of Living Systems - PI
2023 - 27	National Science Foundation , CAREER award - PI
2023 - 28	National Science Foundation , Physics Frontier Center (Center for Living Systems) - Deputy Director
2023 - 26	Chan Zuckerberg Initiative , Theory Initiative - PI
2024 - 29	National Science Foundation + Simons , National Instit. for Theory and Math in Biology - Sr. Personnel
2023 - 27	Sloan Foundation , Matter-to-life - Co-PI
2022 - 27	Moore Foundation , Matter-to-life - Co-PI
2020 - 23	National Science Foundation , UChicago Materials Research Science and Eng Center - Sr. Personnel
2017 - 23	Simons Foundation , Simons Investigator, MMLS - PI

POSTDOCS

2020 -	Riccardo Ravasio	<i>Yen Fellow</i>
2024 -	Mason Rouches	<i>CLS Postdoctoral Fellow</i>
2026 -	Mats van Tongeren	
2025 -	Max Schelling	
2026 -	Cheyne Weis	
2019 - 26	Martin Falk , Currently: Faculty, UMass Amherst (2027)	<i>Schmidt Futures Fellow</i>
2023 - 26	Yoshiya Matsubara , Currently: Research A. Professor, U Tokyo (Jan 2027)	<i>Origins Initiative</i>
2023 - 24	Maryn Carlson , Currently: Faculty, U Rochester (2026)	
2018 - 23	Kabir Husain , Currently: Faculty, Physics, University College London (2024)	<i>James McDonnell Fellow</i>
2017 - 19	Zhiyue Lu , Currently: Faculty, Chemistry, UNC Chapel Hill (2024)	

GRADUATE STUDENTS

2025 -	Serena Debesai	<i>NSF GRFP</i>
2025 -	Ruby Wen	
2024 -	Rio Ondo	<i>PME</i>
2024 -	Kristina Trifonova	<i>NSF GRFP</i>
2023 -	Darren Liu	
2026 -	Kristen Vosberg	
2026 -	Vamsi Varanasi	
2022 -	LaNijah Flagg	
2020 -	Christopher Russo	
2022 - 26	Milena Chakraverti-Wuerthwein	<i>Hertz Fellow</i>
2018 - 25	Eric Rouviere , Currently: Postdoc, JHU (2025)	
2020 - 24	Rudy Mendez Reina	
2017 - 22	Vedant Sachdeva , Currently: Evozyne (biotech startup) (2023)	<i>Harper Dissertation Fellow</i>
2016 - 21	Jackson O'Brien , Currently: Lincoln Park Capital (biotech VC) (2023)	
2016 - 21	Weerapat Pittayakanchit , Currently: Krungthai Bank, Thailand (2023)	
2018 - 20	Chukowunso Arinze , Currently: Kaoshi (founder) (2022)	
2016 - 19	Menachem Stern , Currently: Faculty, AMOLF (2024)	<i>APS GSNP award, Bloomenthal prize</i>

UNDERGRADUATES

HuiYi Mei	<i>2025 - 2026</i>
Leon Zhou	<i>2022 - 2026</i>
Aidan Etterer	<i>2024 - 2025</i>
Michele Peruzzo	<i>2022 - 2025</i>
Sumana Turimella	<i>2022 - 2024</i>
Sedona Kessler , Now: Undergrad, Harvard	<i>2023 - 2024</i>
Shiv Seshan , Now: Grad Student, Caltech (2025)	<i>2023 - 2024</i>
Finn Roach	<i>2021 - 2022</i>
Sophia Smith , Now: Grad Student, UT Austin Applied Math (2022)	<i>2020 - 2021</i>
Sophia Chiang	<i>2020 - 2021</i>
Viraaj Jayaram , Now: Grad Student, Yale Physics (2022)	<i>2017 - 2019</i>
Leron Perez , Now: Grad Student, Stanford (2022)	<i>2018 - 2019</i>
Jon Kutasov , Now: Optiver (fintech) (2021)	<i>2019 - 2019</i>
Amber Bourdier , Now: ESILV, Paris (2019)	<i>2017 - 2018</i>
Ryan Thiermann , Now: Grad Student, UC San Diego Physics (2022)	<i>2015 - 2017</i>
Weishun Zhong , Now: Grad Student, MIT Physics (2022)	<i>2016 - 2017</i>
Alexandra Carruthers Ferrero , Now: Grad Student, Univ Puerto Rico (2018)	<i>2016 - 2016</i>

Invited talks

Sep 26	Cornell , LASSP/AEP Seminar	Ithaca, NY
Sep 26	Annual Origins Federation meeting , Life in the Universe	Zurich, Switzerland
Jul 26	KITP , Learning with a brain	Santa Barbara, CA
Jun 26	Gulbenkian , Molecular and Biophysical Principles in Cellular Evolution	Lisbon, Portugal
Apr 26	Brown , Physics Colloquium	
Mar 26	APS March meeting , AI + Soft Matter Invited Session	Denver, CO
Mar 26	MIT , Biophysics seminar	Cambridge, MA
Feb 26	Georgetown	Washington, DC
Feb 26	Yale , Physics Colloquium	New Haven, CT
Jan 26	Les Houches , Computing with matter	Les Houches, France
Jan 26	Weizmann Institute	Jerusalem, Israel
Jan 26	Tel Aviv University , International Biophysics Prize seminar	Tel Aviv, Israel
Jan 26	Hebrew University	Jerusalem, Israel
Dec 25	American Society for Cell Biology , Cell Behavior and Cognition	Philadelphia, PA
Dec 25	Laufer Center for Physical and Quantitative Biology	Stony Brook, NY
Sep 25	DARPA , Complexity Roundtable	Washington, DC
Jul 25	Molecular Origins of Life 2025	Online
Jul 25	Nordita , Eco-evolutionary Processes in Context	Stockholm, Sweden
Jun 25	ESPCI , Chaire Joliot seminar	Paris, France
Jun 25	Beg Rohu Summer school , Learning with Machines, Physics and Minds	Beg Rohu, France
May 25	Aspen Colloquium , Beyond von Neumann computing	Aspen, CO
May 25	NITMB , Origin of Life and Self-Organized Complexity	Chicago, IL
May 25	EMBL , Theory Transversal Theme retreat	Heidelberg, Germany
May 25	EMBL Symposium , Theory and concepts in biology	Heidelberg, Germany
Apr 25	Johns Hopkins , Materials Science Seminar	Baltimore, MD
Jan 25	UC Berkeley , Annual Berkeley Stat Mech Meeting	Berkeley, CA
Jan 25	National Institute for Theory and Math in Biology , Biological systems that learn	Chicago, IL
Nov 24	Cornell , Soft Matter Hub	Online
Aug 24	Aspen Summer Workshop Data in Biology , Biological Data Analysis	Aspen, CO
Jul 24	MOLINFO 2024 , Molecular Information	Munich, Germany
Jul 24	University of Tokyo , Workshop on Frontiers in Biophysics	Tokyo, Japan
Jun 24	IUPAB congress 2024 , Theory in Biology symposium	Kyoto, Japan
Jun 24	Gordon Research Conference , Systems Chemistry	Portland, ME
Jun 24	University of Pennsylvania , Center for Soft and Living Matter kickoff	Philadelphia, PA
May 24	MIT , Applied Math Seminar	Cambridge, MA
Apr 24	1st Solvay conference in Biology , The organisation and dynamics of biological computation	Brussels, Belgium
Apr 24	Simons Foundation , NITMB Annual Meeting	New York, NY
Mar 24	APS March meeting , Invited talk	Minneapolis, MN
Jan 24	Aspen Institute for Physics , Computation in Physical systems	Aspen, CO
Oct 23	29th Solvay conference on Physics , Structure and Dynamics of Disordered Systems	Brussels, Belgium
Sep 23	American Academy of Arts and Sciences , Life in the Universe II	Boston, MA
Sep 23	University of California, Merced (Remote) , Physics colloquium	Merced, CA
Jul 23	University of Michigan , ICAM Complex Mechanical Metamaterials workshop	Ann Arbor, MI
Jun 23	Gordon Conference on Molecular Mechanisms of Evolution	Stonehill College, MA
Jun 23	Northwestern University , qBio Symposium	Chicago, IL
Mar 23	APS March meeting , Statistical physics Tutorial	Las Vegas, NV
Mar 23	NYU , Physics colloquium	NYC, NY
Feb 23	Harvard Medical School , Theory lunch	Boston, MA
Jan 23	Gordon Research Conference , Stochastic Processes in Biology	Ventura, CA
Nov 22	AMOLF , Physics Colloquium	Amsterdam, NL
Nov 22	AMOLF , Workshop on Information Processing	Amsterdam, NL
Nov 22	TU Eindhoven , Physics/Chemistry Colloquium	Eindhoven, NL
Oct 22	Stanford , Coherent Network Computing	Palo Alto, CA
Sep 22	Cargese Summer School , Summer school	Corsica, France
May 22	Simons Center for Geometry & Physics , Geometry, Topology, Symmetry in Soft & Living Matter	Stony Brook, NY
May 22	Johns Hopkins , Biophysics seminar	Baltimore, MD
Apr 22	Harvard University , Widely Applied Mathematics	Cambridge, MA

Apr 22	Illinois Institute of Technology	Chicago, IL
Mar 22	APS March Meeting , Invited talk	Chicago, IL
Sep 21	University of Michigan , Physics seminar	Ann Arbor, MI
May 21	Cold Spring Harbor (remote)	Cold Spring Harbor, NY
May 21	University of Amsterdam (remote)	Amsterdam, NL
Apr 21	Biological Physics/Physical Biology series (Virtual)	Zoom
Jan 21	qEvo 2021	Zoom
Dec 20	Max Planck Institute (remote)	Gottingen, Germany
Nov 20	Imperial College , Physics of life series	London, UK
Jun 20	Tel Aviv University , Workshop on Pathways, Sequence and Memory	Tel Aviv, Israel
Feb 20	Banff International Research Station , Workshop on gene regulation	Banff, Canada
Jan 20	Massachusetts Institute of Technology , Physics seminar	Cambridge, MA
Jan 20	EPFL , CECAM workshop	Lausanne, Switzerland
Oct 19	University of California, Los Angeles , Physics seminar	Los Angeles, CA
Sep 19	National Science Foundation , MRSEC Directors meeting	Alexandria, VA
Sep 19	University of Pennsylvania , Physics seminar	Philadelphia, PA
Jul 19	Aspen Center for Physics , Information processing in cells	Aspen, CO
Jun 19	Frontiers of Biophysics	Paros, Greece
Mar 19	Chan-Zuckerberg Biohub , Theory in Biology day	San Francisco, CA
Jan 19	Gordon Research Conference , Stochastic Physics and Biology	Ventura, CA
Nov 18	University of Illinois , Physics seminar	Urbana-Champaign, IL
Oct 18	Peking University , Center for Quantitative Biology	Beijing, China
Oct 18	DNA 24 , Plenary Talk	Dajin, China
Oct 18	Penn State , Physics seminar	College Station, PA
Apr 18	Emory University , Physics Colloquium	Atlanta, GA
Apr 18	Simons Foundation , Theory in Biology conference	New York, NY
Apr 18	Rice University , Physics seminar	Houston, TX
Mar 18	APS March Meeting , Invited talk	Los Angeles, CA
Feb 18	KITP , Memories workshop	Santa Barbara, CA
Nov 17	Princeton Center for Theoretical Science , Workshop on frustration	Princeton, NJ
Oct 17	MIT , Physics seminar	Cambridge, MA
Oct 17	University of Texas Southwestern , Systems Biology Seminar	Dallas, TX
Sep 17	Washington University in St. Louis , Physics seminar	St Louis, MO
Jul 17	QBio workshop	New Brunswick, NJ
May 17	Los Alamos National Lab , CNLS Annual workshop	Santa Fe, NM
Mar 17	Harvard University , Widely Applied Math seminar	Cambridge, MA
Jan 17	Northwestern University , Mechanical Engineering,	Chicago, IL
May 16	Harvard University , Workshop on Aging and Failure in biology	Cambridge, MA
Apr 16	Foundations of Nanoscience (FNANO) , Invited speaker	Snowbird, UT
Mar 16	University of Illinois Urbana-Champaign , Urbana cond-mat Symposium	Urbana, IL
Sep 15	Santa Fe Institute , Kinetic Networks workshop	Santa Fe, NM
Apr 15	Rockefeller University , Center for Studies in Physics and Biology	New York, NY
Mar 15	University of Colorado , Physics Colloquium	Boulder, CO
Mar 15	University of Colorado , Physics seminar	Boulder, CO
Mar 15	APS March Meeting , Invited Talk	San Antonio, TX
Mar 15	University of Massachusetts , Physics seminar	Amherst, MA
Feb 15	University of San Diego , Physics seminar	San Diego, CA
Feb 15	MIT , Physics seminar	Cambridge, MA
Feb 15	Aspen Institute for , Unified Concepts in Glass Physics VI,	Aspen, CO
Jan 15	Cornell , LASSP Applied Physics seminar	Ithaca, NY
Jan 15	University of Chicago , James Franck Institute,	Chicago, IL
Dec 14	Princeton University , Department of Chemistry,	Princeton, NJ
Jul 14	Universite Joseph Fourier	Grenoble, France
Jun 14	University of Nice , Laboratory of Condensed Matter Physics seminar	Nice, France
Jun 14	ESPCI	Paris, France
Jun 14	Brandeis , Physics Department	Waltham, MA
May 14	University of Chicago , Computations in Science Seminar	Chicago, IL
Apr 14	University of Maryland , Institute for Physical Science and Technology,	College Park, MD
Jan 14	Northwestern , Applied Math Colloquium	Evanston, IL

Dec 13	Boston Physical Biology Hangout	<i>Cambridge, MA</i>
Nov 13	Harvard , Bauer Forum	<i>Cambridge, MA</i>
Dec 08	Perimeter Institute for Theoretical Physics , Young Researchers Conference	<i>Waterloo, ON</i>
May 08	Institute for Nuclear Theory , String Theory Methods in the Real World,	<i>Seattle, WA</i>
Apr 08	Princeton University , High-energy theory group seminar	<i>Princeton, NJ</i>
Apr 08	Cornell University , High-energy theory group seminar	<i>Ithaca, NY</i>
Nov 07	University of Chicago , High-energy theory group seminar	<i>Chicago, IL</i>